

Design a Web Crawler - Workflow

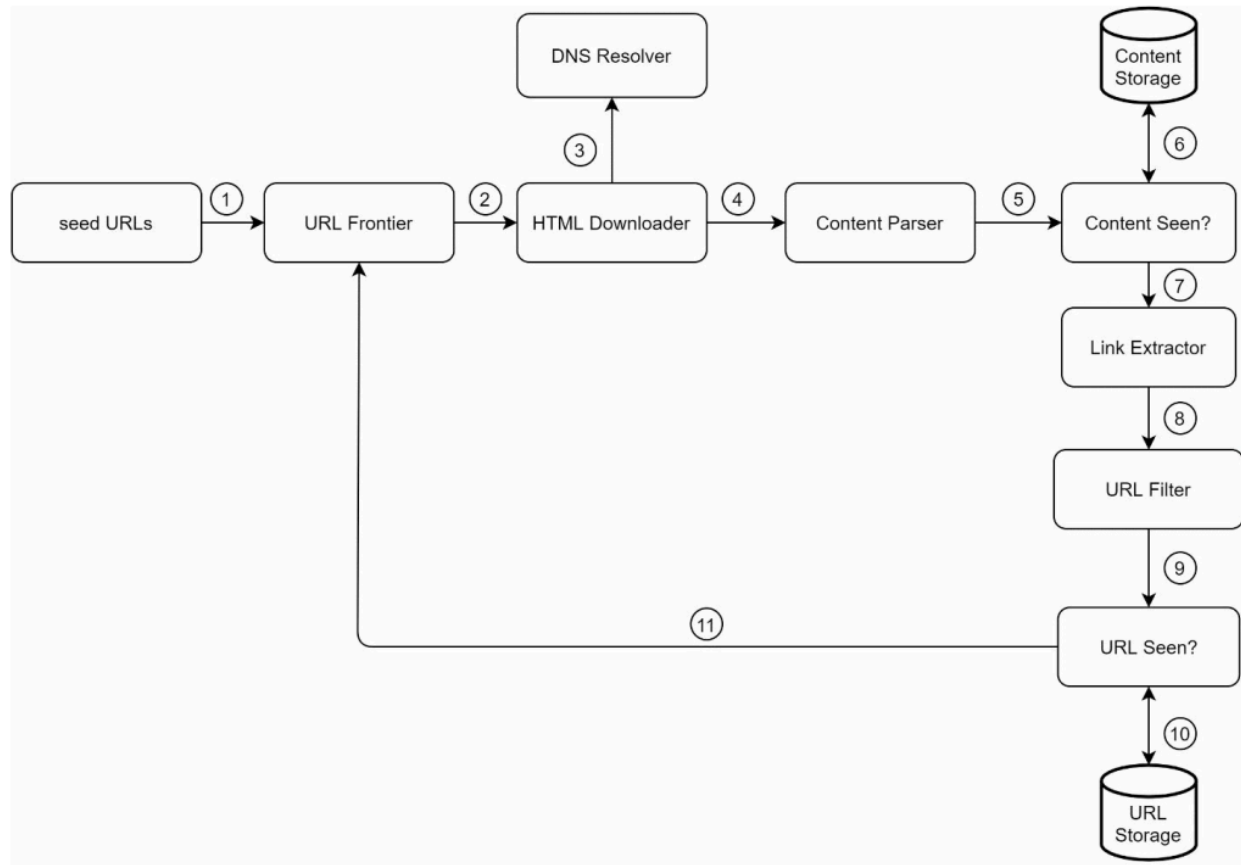
1. Overview

The web crawler workflow defines how the system:

- Collects web pages
- Extracts new URLs
- Avoids duplication
- Continuously expands crawling scope

The process is iterative and scalable.

2. Step-by-Step Workflow



Step 1: Seed URLs Initialization

- Initial set of URLs (called **seed URLs**) are added to the system
- These act as the **starting point** for crawling

Stored in **URL Frontier (queue)**

Step 2: Fetch URLs from Frontier

- HTML Downloader retrieves a batch of URLs from the **URL Frontier**
- Ensures continuous crawling pipeline

Step 3: DNS Resolution + Download

- For each URL:
 - DNS Resolver converts URL → IP address
 - HTML Downloader sends request and downloads web page

Step 4: Content Parsing

- Downloaded HTML pages are sent to **Content Parser**
- Parser:
 - Validates HTML structure
 - Removes malformed or invalid pages

Step 5: Duplicate Content Check (Content Seen?)

- Parsed content is checked against stored content

Cases:

- **Already exists**
 - Duplicate content → discard
- **New content**
 - Forward to next stage

Uses hashing for efficient comparison

Step 6: Link Extraction

- URL Extractor parses HTML and extracts all links

Converts:

- Relative URLs → Absolute URLs

Step 7: URL Filtering

- Extracted URLs are filtered

Removes:

- Invalid URLs
- Blocked/blacklisted domains
- Unsupported file types

Step 8: URL Deduplication (URL Seen?)

- Check if URL was already processed or exists in queue

Cases:

- Already seen → Ignore
- New URL → Process further

Implemented using:

- Hash tables / Bloom filters

Step 9: Add New URLs to Frontier

- New unique URLs are added back to **URL Frontier**

Enables continuous crawling loop

3. End-to-End Flow Summary

None

Seed URLs



URL Frontier (Queue)



HTML Downloader



DNS Resolver



Content Parser



Content Seen? (Duplicate Check)



URL Extractor



URL Filter



URL Seen? (Deduplication)



URL Frontier (Loop Continues)

4. Key Observations

- System is **iterative and cyclic**
- Avoids:
 - Duplicate content
 - Duplicate URLs
- Ensures:
 - Scalability
 - Efficiency

- Continuous discovery

5. Key Takeaway

Web crawler workflow:

- Continuously **download** → **process** → **extract** → **filter** → **repeat**
- Uses multiple layers of **deduplication and validation**
- Designed for **large-scale parallel crawling**